Session Objectives

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1. Describe the phenomena of tolerance in immunity and the pathologic implications
2. Compare central to peripheral tolerance
3. Define autoimmunity
4. Describe the three stages of an autoimmune disease
5. Identify the mechanisms of pathogenesis of autoimmune diseases

Source: Course Syllabus

Tolerance | Lymphocyte Receptor

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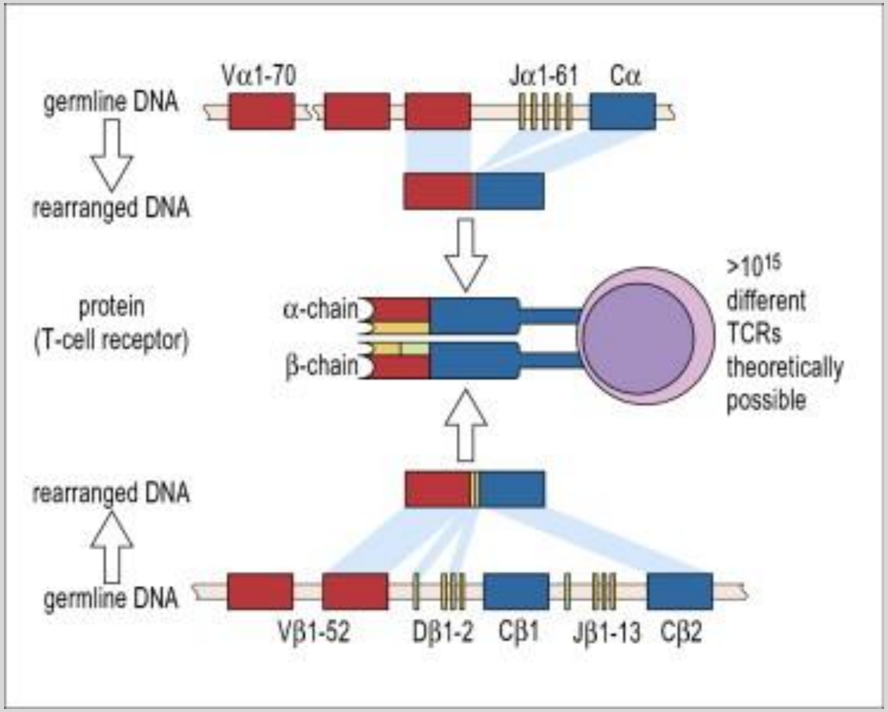
- **Fulfill two contradictory requirements:**

  1. Large variety of different receptors to avoid 'holes' that could be exploited by pathogens

  2. the receptor repertoire must be shaped to prevent the immune system from attacking the host

Generation of T–cell receptor (TCR)



Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3 figure 19.1.

Origins of Autoimmunity

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- Paul Ehrlich coined the term ***horror autotoxicus*** for the necessity to avoid immunological reactions against self-antigens

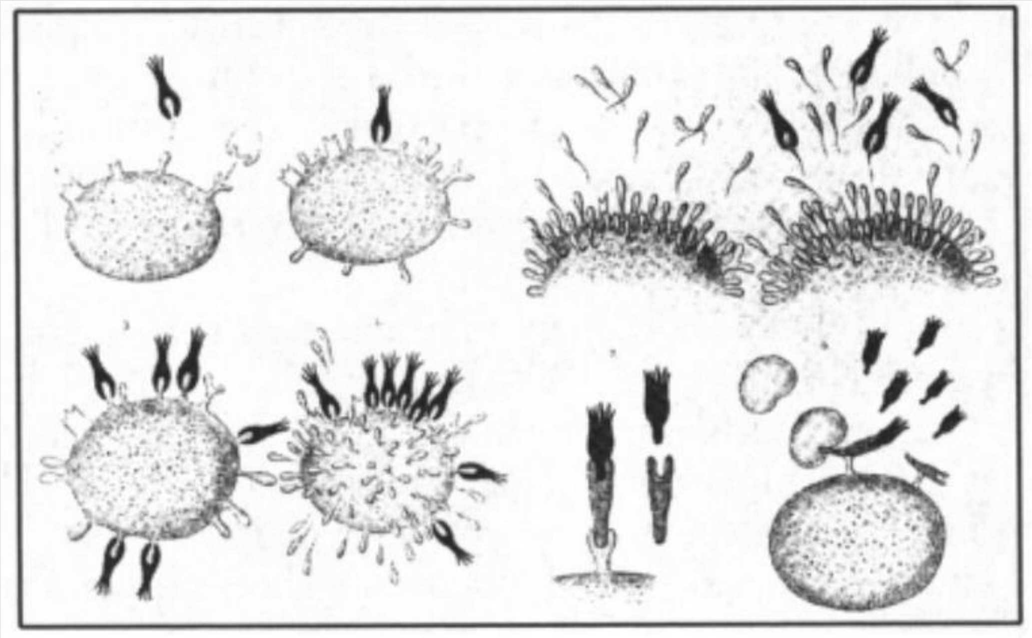

Illustration of Ehrlich's Side Chain Theory

Source: Travis AS. *History and Philosophy of the Life Sciences* Vol. 30, No. 1 (2008), pp. 79-97

- **Antigen-specific unresponsiveness**
- Result of a programmed developmental process
- Adaptive IR learns to tolerate:
  1. self-antigens (autoantigens)
  2. exogenous antigens (dietary proteins)
- Autoimmunity requires the presence of self-reactive CD4 $^{+}$ T-cells (*exception*)

Central Tolerance | T-cells

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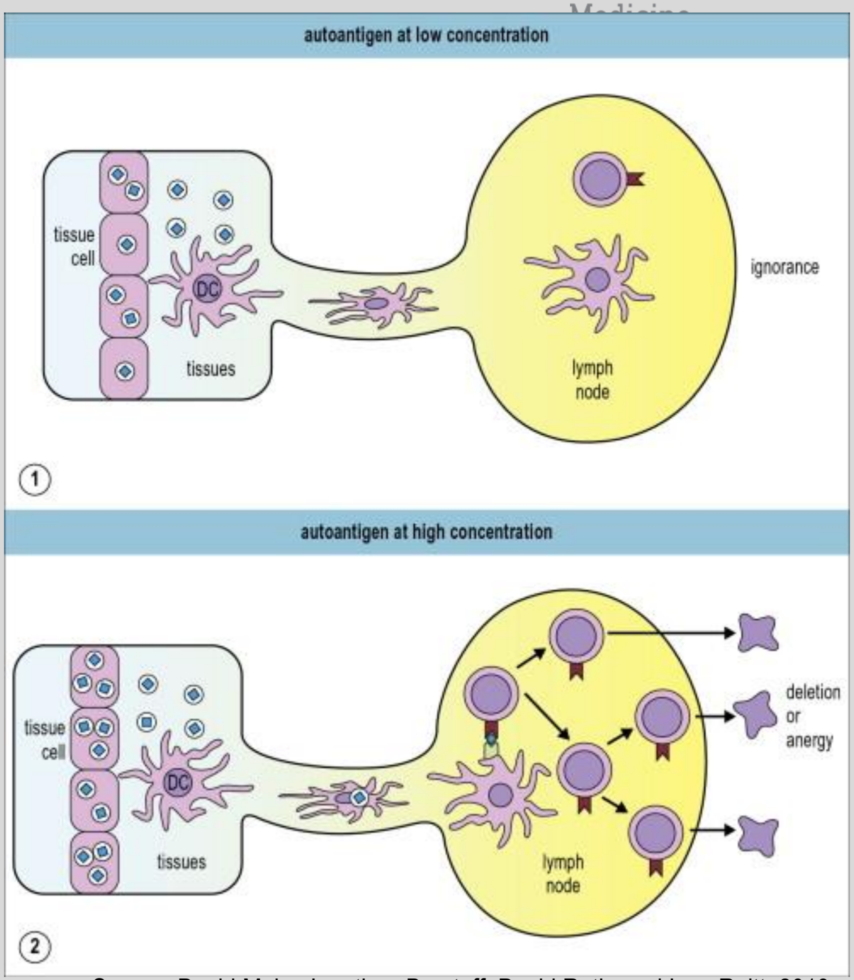
1. 30% of low-avidity self-reactive T-cells clones → periphery
2. Thymic deletion of autoreactive T-cells is not the only mechanism responsible for self-tolerance → peripheral tolerance
3. Antigen dose and TCR avidity
4. Medium to high affinity self reactive T-cells become T_{reg}

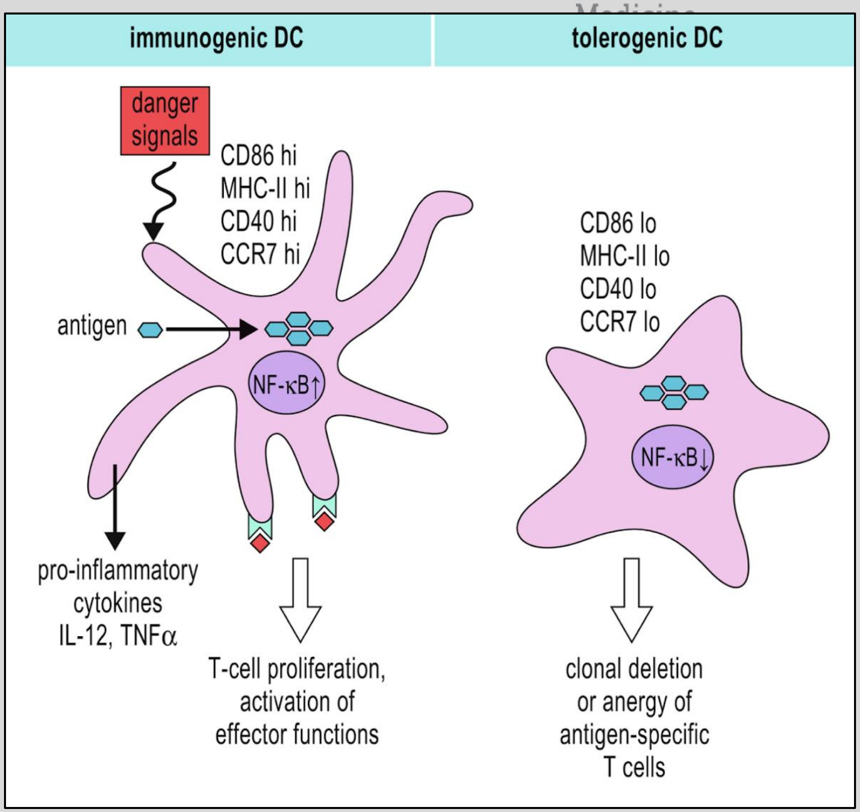


Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. Immunology 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig 19.w1

Source: Kondělková K, Vokurková D, Krejsek J, Borská L, Fiala Z, Ctirad A. Regulatory T cells (TREG) and their roles in immune system with respect to immunopathological disorders. Acta Medica (Hradec Kralove). 2010;53(2):73-7. doi: 10.14712/18059694.2016.63. PMID: 20672742.

Central Tolerance | T-cells

1. Autoreactive T-cells are part of the **normal repertoire** (*e.g., T-cells that recognize insulin or myelin can be isolated from people without diabetes or MS*)
2. **Tolerogenic Dendritic cells** (*not immunogenic*)



Source: Figure 11.7 from Chapter 11 in D. Male, R. Stokes Peebles, and V. Male. 2021. ***Immunology*** 9th edition. Elsevier publisher. ISBN: 978-0-7020-7844-6

Peripheral Tolerance | T_{reg} cells

Medium → high affinity self reactive T-cells

FOXP3: regulatory gene for development and function

T_{reg} cell

function

Prevent and suppress immune responses

Cytotoxic to:
- -CD8+ T cells
- -NK cells
- -B-cells
- -DCs

Secretion of

IL-10

TGF-β

IL-35

Depletion of IL-2

identity

10% of all CD4+ T-cells

Selectively express CD3 receptor

Importance:

- Dampen responses against microbial antigens, allergens, tumors, & allografts, and protect fetuses during pregnancy
- IPEX (immunodysregulation, polyendocrinopathy, enteropathy, X-linked) syndrome
- Autoimmunity due to loss of function or effectiveness (effector T cells become resistant to their action)

Review Questions

1. Which of the following is unique to central B-cell tolerance?
   A. Immunologic Ignorance
   B. Receptor Editing
   C. Anergy
   D. Apoptosis

2. Which of the following would not be used as a marker for T_{reg} cells?
   A. CD25+
   B. FOXP3
   C. CD3+
   D. CD8+

3. A mutation in the FOXP3 gene would directly affect which of the following cell types?
   A. T-helper cells
   B. NK cells
   C. T-regulatory cells
   D. Dendritic cells

Definition of Autoimmunity

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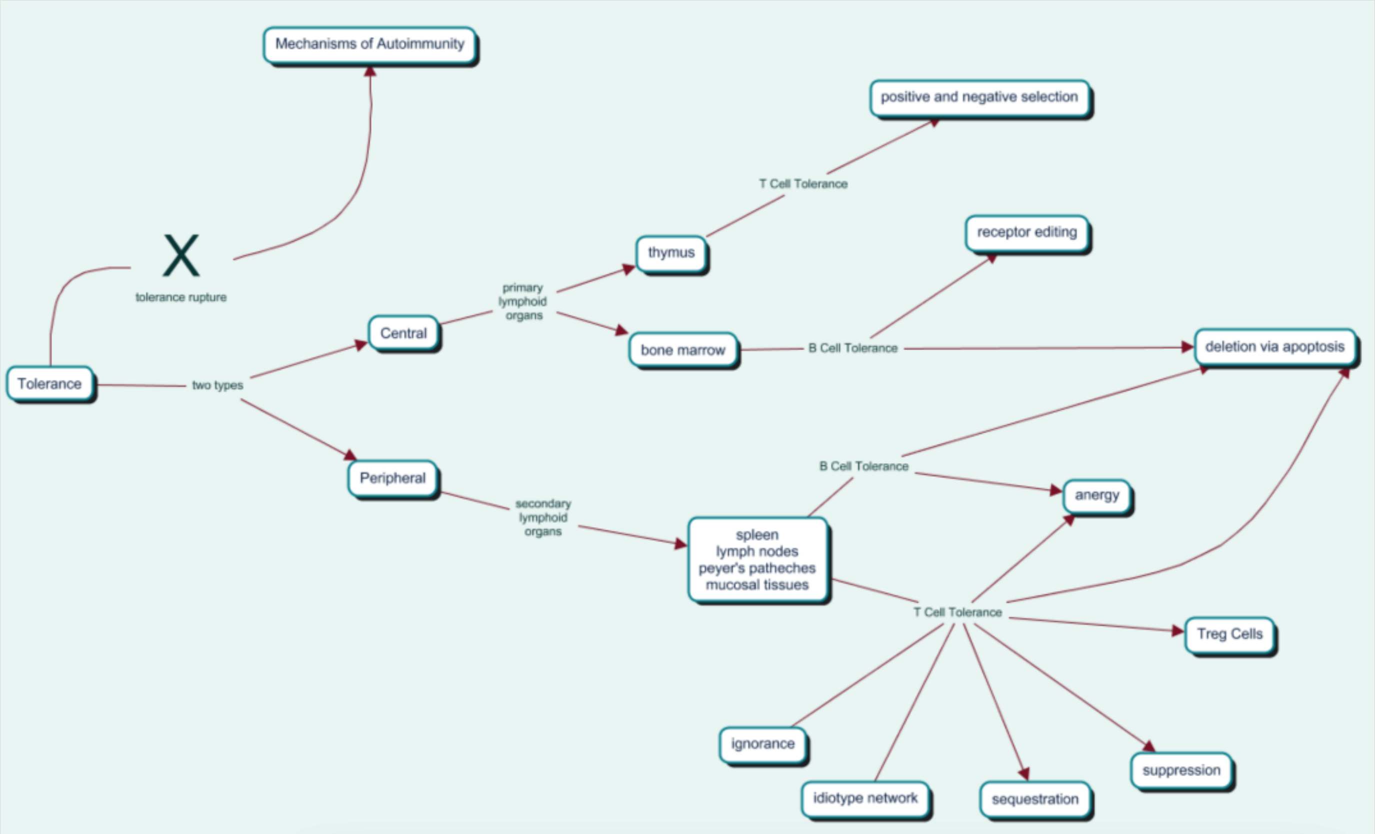
- Autoimmunity occurs when:
  1. Self-tolerance broken (*tolerance rupture)* and an individual becomes victim of their own immune response
  2. Loss of previously established tolerance



Autoimmunity

Antigen driven process characterized by:

1. Specificity
2. High affinity
3. Memory

+

Autoreactive lymphocyte or autoantibody

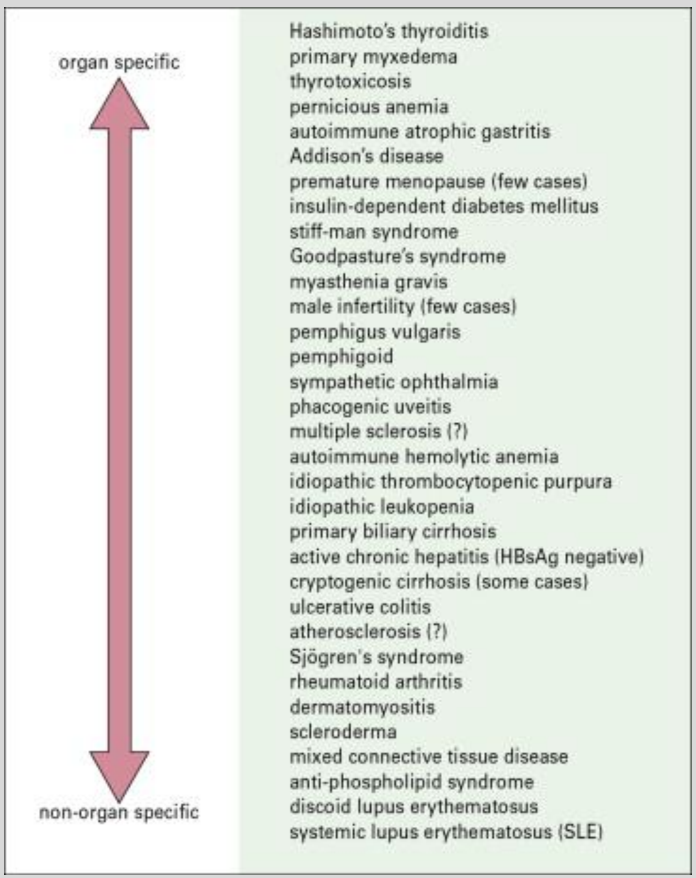
→

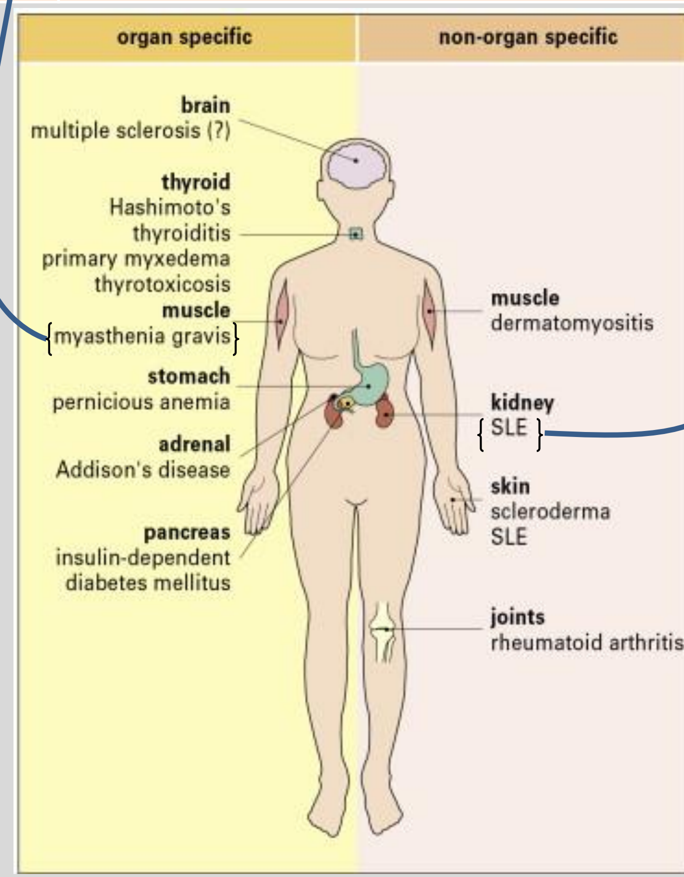
pathogenesis

recognition of an antigen by the IS (either foreign or self) → assault on its own self-antigens

Autoimmune Diseases

Clinical Categories	Organ Specific	Systemic
Mechanism	Serum reacts to single autoantigen/restricted group of autoantigens in one organ	Serum reacts to multiple autoantigens





Source: Male D, Brostoff J, Roth D, and Roitt I (2013). Immunology 8th ed. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig. 20.4 and 20.3

Autoimmune Diseases

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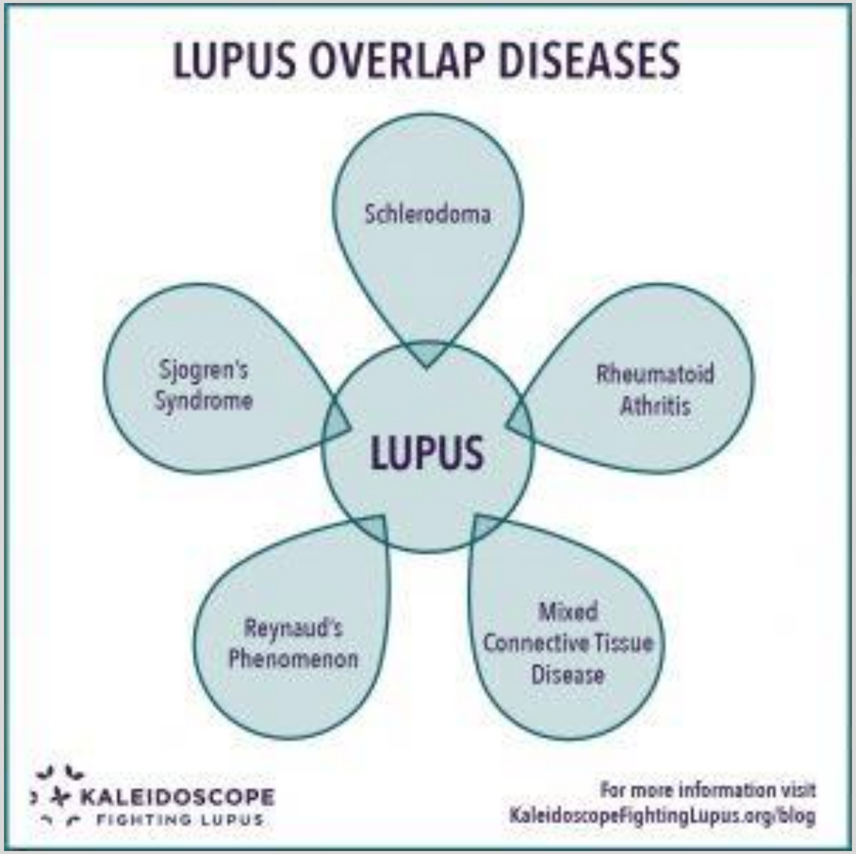
- Autoimmune diseases ‘hunt in packs’
  - 30% of patients with SLE & Sjögren's syndrome have a second, third, or even fourth autoimmune disease

- Systemic autoimmune rheumatic diseases show considerable overlap



Source: Heintz L. Lupus Overlap Diseases and Overlap Syndrome. https://kaleidoscopefightinglupus.org/lupus-overlap-diseases-and-overlap-syndrome/

Stages of Autoimmune Diseases

Three Stages

Priming

Subclinical

Clinical

Source: https://onlinelibrary.wiley.com/doi/full/10.1002/eji.201444486

Stages of Autoimmune Diseases

Stage One: Priming

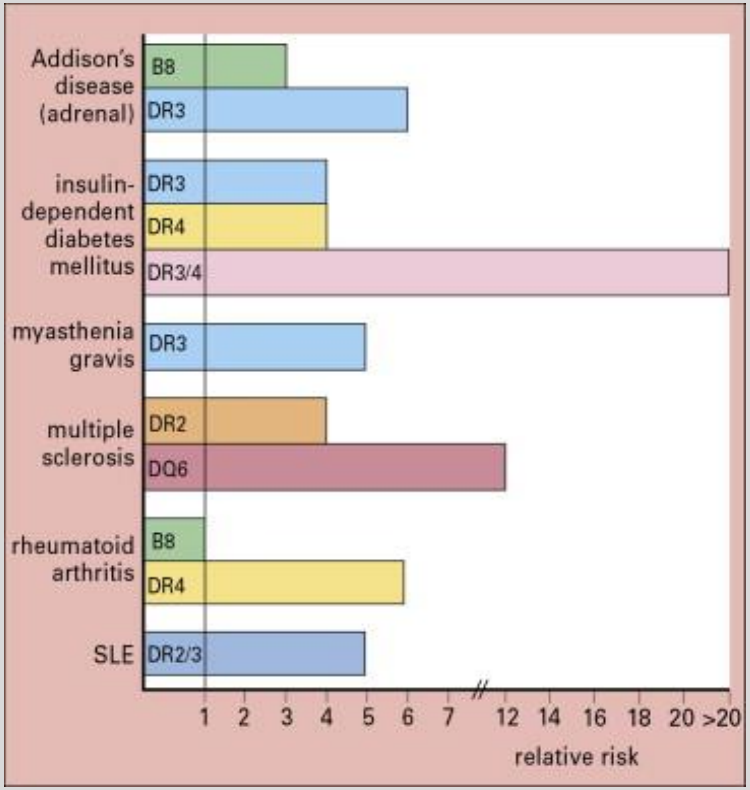
Genetic

genes that confer a genetic risk

polygenic; HLA-B27

Environmental Exposure

Infection or UV radiation



Source: Male D, Brostoff J, Roth D, and Roitt I (2013). Immunology 8th ed. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig. 20.6

Review Questions

1. A certain autoimmune condition causes inflammation, vasculitis, and perifascicular atrophy. Is this an organ specific or systemic autoimmune disease? _______________.

2. Which of the following is true of autoimmunity?
   A. It involves the loss of tolerance
   B. It occurs in any individual with autoreactive T-cells
   C. An individual can only have one autoimmune disease
   D. There are no known genetic risk factors

Mechanism of Pathogenesis of Autoimmune Diseases

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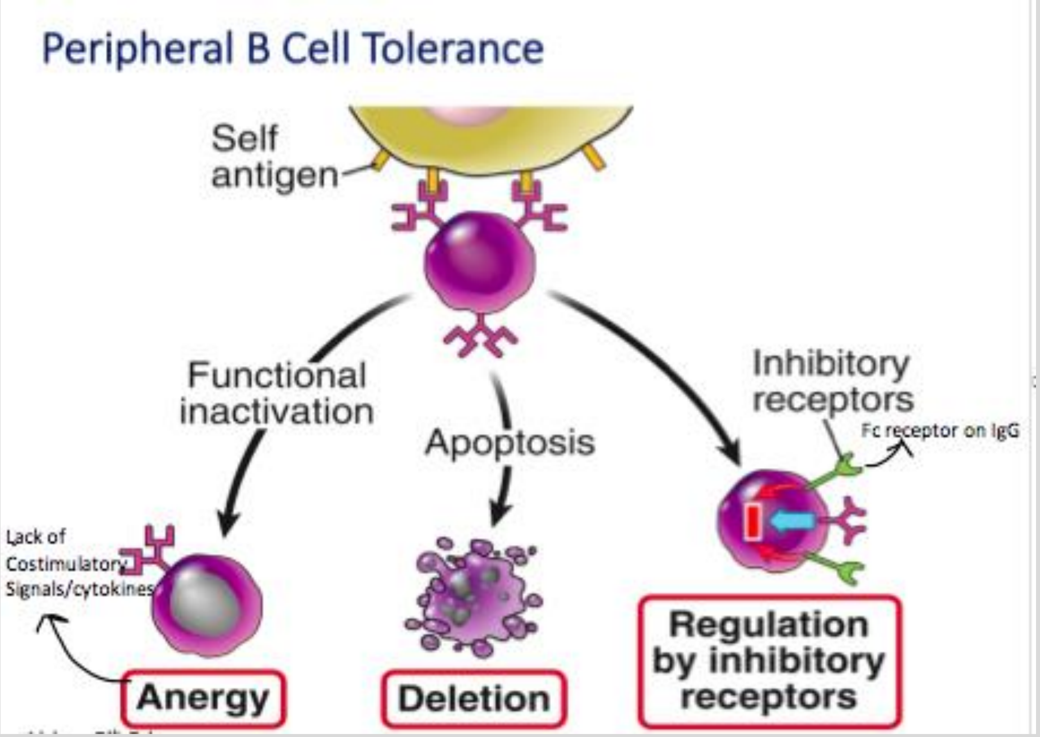
- Super Antigens
- Defective Receptors
- Molecular Mimicry
- Defective Apoptosis
- Epitope Spreading
- Cryptic Epitope Exposure
- Hapten Effect
- Inappropriate MHC Expression and Upregulation
- Defective T_{reg} Response
- Hormonal Imbalances
- Cytokine Dysregulation
- Pre-Existing Organ Defects

- **Autoreactive B-cells with defective receptors**
  - o do not respond to inhibitory signals to maintain self-tolerance



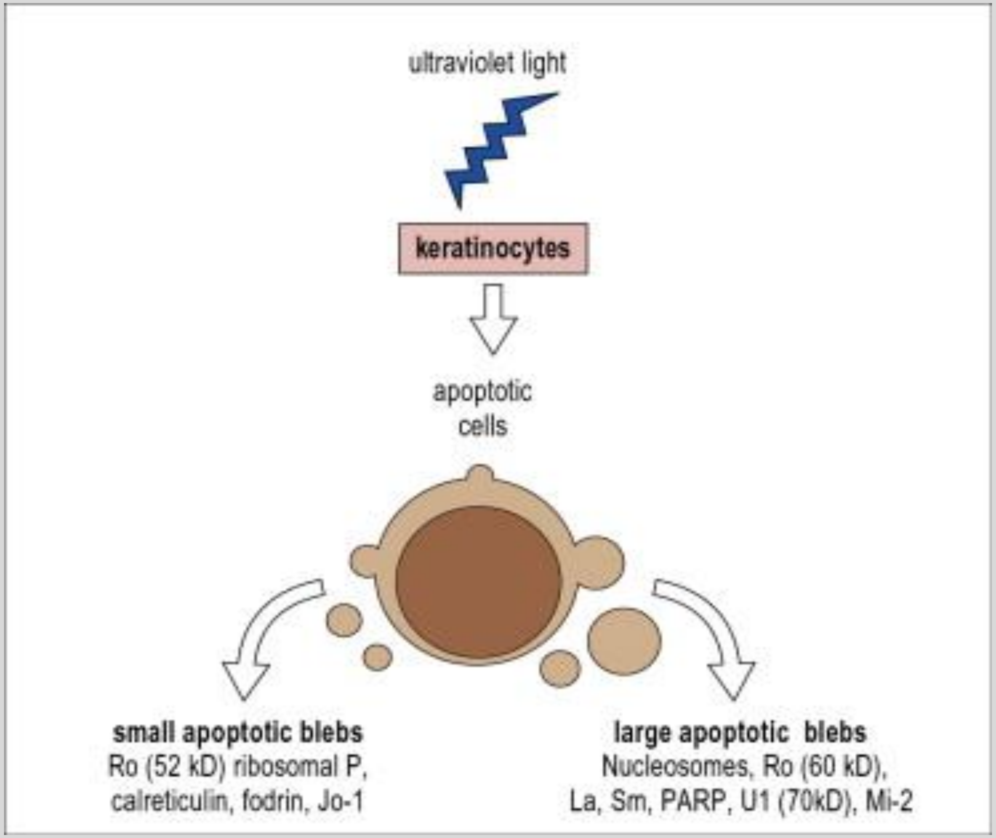
Source: https://memorang.com/flashcards/146143/Lecture+4+-+Autoimmunity

Defective Apoptosis:

a) Accelerated apoptosis of cells with increased release of self-antigens

b) Defective presentation of apoptotic cell antigens to lymphocytes

c) Failure of clearance of apoptotic cells by Mϕ → prolonged exposure of internal autoAgs (SLE)



Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. Immunology 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig 20.10

Cytokine Dysregulation

excessive cytokine production

or

defective cytokine receptors

aberrant immune response and/or activation of anergic self-reactive T-cells

(IL2RA; TNF-, INFγ and B-cell destruction)

- **Pre-existing defects in the target organ:**
  a) Increase uptake of iodine by thyroid
  b) Risk factors linked to a TF controlling the rate of insulin production in type I diabetes

Summary

- Tolerance is a complex immunological process that occurs in primary and secondary lymphoid organs
- Tolerance rupture → autoimmunity
- Autoimmunity: 2 clinical categories
- Autoimmune disease involves 3 stages
- Mechanisms/ongoing research